L6

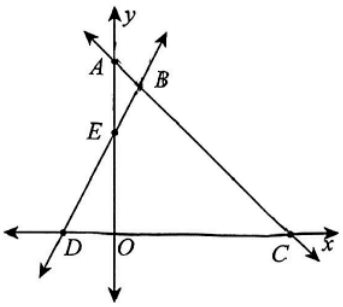
3. Given that $y = (a+1)x^{a^2} + (b-2)$ is a direct variation with respect to x, then $(a-b)^{2017} =$ _____ .

4. There is an equation $y = (m-1)x^{|m|} + n - 4$.

(1) If this is a linear equation, what is the value of m and n?

9. A line with slope 2 intersects a line with slope 6 at the point $(40, 30)$. What is the distance between the x-intercepts of these two lines? _____ . (Adapted From 2018 AMC 12B Problems, Question #3)

10 As shown in the figure, line $l_1 : y = 2x + 4$ intersects with $l_2 : y = -x + 7$ at point B. Line l_1 intersects with the x-axis at point D and with the y-axis at point E; line l_2 intersects with the x-axis at point C and with the y-axis at point A. The area of the quadrilateral $OCBE$ is _____



L7

2 After moving the line $y = -x + 3$ upward by m units, the y-coordinate of the point of intersection with the line $y = 2x + 4$ is 6, then $m =$ _____ .

5 There are two linear equations $y_1 = ax + 2$ and $y_2 = 3x - b$. Answer the following questions.

(3) If y_1 and y_2 are symmetric about the origin, $a =$ _____ , $b =$ _____ .

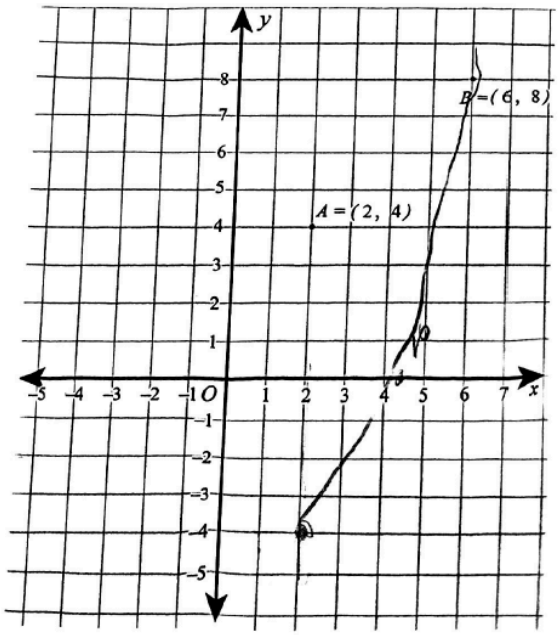
8 If the graph of the line $y = 2x - 1$ is rotated $90°$ counterclockwise about the origin O, the equation of the line after the rotation is _____ .

L7 Extention

7 The line l and the line $y = -3x - 2$ are symmetric about the x-axis. Then the expression of the line l is _____ .

A. $y = 3x - 2$ B. $y = \frac{1}{3}x + 2$ C. $y = -3x + 2$ D. $y = 3x + 2$

8 There are two points $A\ (2, 4)$ and $B\ (6, 8)$ on the coordinate plane. Point P is on the x-axis. If $PA + PB$ is at the minimum value, what are the coordinates of point P?



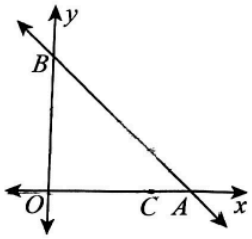
5 The line that is symmetric to line $y = mx - 3$ about the x-axis passes through $(-1, 2)$, then the value of m is _____ .

A. 1 B. -1 C. 5 D. -5

8 The equation of the line obtained by rotating line $y = 3x$ clockwise around the origin $90°$ is _____ ; The equation of the line obtained by rotating line $y = 3x$ counterclockwise around the origin $90°$ is _____

9 If line $y = 2x - 3$ is rotated $90°$ clockwise about the origin O, the equation of the line after rotation is _____ ; if the line is rotated $90°$ counterclockwise about the origin O, the equation of the line after rotation is _____ .

10 The graph of the linear equation $y = -x + 5$ intersects with x-axis and y-axis at A and B respectively .



(2) Given that the coordinates of point C is $(4,0)$. Suppose point D is symmetric to C with respect to the line AB. Please write down the coordinates of point D directly.

(3) Points M and N are on the line AB and y-axis respectively, so that the perimeter of $\triangle CMN$ is at its minimum. What are the coordinates of point N?

L8

(2) $a^2 + b^2 + c^2 + ab + bc - ca =$ _____ .

(3) $a^2 + b^2 + c^2 - ab + bc - ca =$ _____ .

3 Calculate the following:

(1) $(2+1)(2^2+1)(2^4+1)\cdots(2^{2^n}+1) =$ _____ .

(2) $(a+b+c-d)^2+(a+b-c+d)^2+(a-b+c+d)^2+(-a+b+c+d)^2=$ _____ .

(3) $(2a+b)^2\left[4a^2-(2a-b)b\right]^2=$ _____

5 If $a+b=5$, find the value of a^3+b^3+15ab.

6 Answer the following questions:

(1) If a, b, c satisfy the equation $\begin{cases} b^2 + 2ac = 14 \\ c^2 + 2ab = 29 \\ a^2 + 2bc = 21 \end{cases}$, then the value of $a + b + c$ is _____ .

(2) Given that $(x + y)^2 - 2x - 2y + 1 = 0$, then $(x + y)^{999} =$ _____ .

8 Calculate: $\dfrac{1999^3 - 1000^3 - 999^3}{1999 \times 1000 \times 999}$

L8 Extention

7 Calculate:

(1) $6(7+1)(7^2+1)(7^4+1)(7^8+1)+1=$ _____ .

6 Calculate:

(1) $(2y-x-3z)(-x-2y-3z)=$ _____ .

10 Which of the following is equivalent to
$(2+3)(2^2+3^2)(2^4+3^4)(2^8+3^8)(2^{16}+3^{16})(2^{32}+3^{32})(2^{64}+3^{64})$?

(2021 Spring AMC 10A Problems, Question #10)

A. $3^{127}+2^{127}$

B. $3^{127}+2^{127}+2\cdot 3^{63}+3\cdot 2^{63}$

C. $3^{128}-2^{128}$

D. $3^{128}+2^{128}$

E. 5^{127}